

1) [4pts] Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a transformation. Which two of the following statements are equivalent to the statement " T is an *onto* transformation"? Circle your answers.

- a) Each element in $\mathbb{R}^n$ has only one image in $\mathbb{R}^m$.
- b) Each element in $\mathbb{R}^n$ maps into some element in $\mathbb{R}^m$.
- c) Each element in $\mathbb{R}^m$ is the image of at most one element in $\mathbb{R}^n$.
- ☒ d) Each element in $\mathbb{R}^m$ is the image of at least one element in $\mathbb{R}^n$.
- e) For every $\mathbf{x}_1$ and $\mathbf{x}_2$ in $\mathbb{R}^n$, $\mathbf{x}_1 \neq \mathbf{x}_2$ implies $T(\mathbf{x}_1) \neq T(\mathbf{x}_2)$.
- ☒ f) The equation $T(\mathbf{x}) = \mathbf{b}$ has a solution for some $\mathbf{b}$ in $\mathbb{R}^m$.
- ☒ g) The equation $T(\mathbf{x}) = \mathbf{b}$ has a solution for every $\mathbf{b}$ in $\mathbb{R}^m$.
- h) For every $\mathbf{b}$ in $\mathbb{R}^m$, the equation $T(\mathbf{x}) = \mathbf{b}$ has a unique solution or none at all.

2) For parts (a) to (e), let P and Q be 3×3 matrices. Given $\det(P^{-1}) = 12$ and $\det(Q) = \frac{1}{2}$, find each of the following determinants.

a) [3pts] $\det(P) = \underline{\frac{1}{12}}$

$$\frac{1}{\det(P^{-1})} = \frac{1}{12}$$

b) [3pts] $\det(Q^T) = \underline{\frac{1}{2}}$

$$\det(Q^T) = \det(Q)$$

c) [3pts] $\det(2P^{-1}) = \underline{96}$

$$\det(2P^{-1}) = 2^3 \det(P^{-1}) = 8(12) = 96$$

d) [3pts] $\det(P^{-1}Q) = \underline{6}$

$$\det(P^{-1}Q) = \det(P^{-1}) \det(Q) = (12)\left(\frac{1}{2}\right) = 6$$

e) [6pts] If $Q \xrightarrow{R_2 \rightarrow R_2 - R_1} Q_1 \xrightarrow{R_3 \rightarrow 10R_3} Q_2 \xrightarrow{R_2 \leftrightarrow R_3} Q_3 \xrightarrow{R_1 \rightarrow R_1 + R_3} Q_4$, then $\det(Q_4) = \underline{-5}$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \frac{1}{2} \end{bmatrix} \xrightarrow{R_2 - R_1} \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & \frac{1}{2} \end{bmatrix} \xrightarrow{10R_3} \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 5 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 5 \\ -1 & 1 & 0 \end{bmatrix}$$

$$\begin{matrix} R_1 + R_3 \\ R_2 - 2R_3 \end{matrix} \rightarrow \begin{bmatrix} 0 & 1 & 0 \\ 2 & -2 & 5 \\ -1 & 1 & 0 \end{bmatrix} \Rightarrow \det(Q_4) = 0 \begin{vmatrix} 1 & 0 \\ -1 & 0 \end{vmatrix} - 1 \begin{vmatrix} 2 & 5 \\ -1 & 0 \end{vmatrix} + 0 \begin{vmatrix} 2 & -2 \\ -1 & 1 \end{vmatrix}$$

$$\det(Q_4) = -1(0 + 5) = -5$$

Do not write below this line (five points off).

problem	1	2	3	4	5	6	7	8	9	10	Σ
earned	4	16	4	10	7	5	10	7	6	10	79
possible	4	18	8	10	14	8	10	8	10	10	100

3) [8pts] Let T be a linear transformation and A be its standard matrix.

Fill in the table with either "Yes" or "No". No explanation necessary.

Size of matrix A	Number of pivot positions of A	Is T onto?	Is T one-to-one?
5×5	5	Yes ✓	Yes ✓
5×4	4	Yes	No
4×5	3	No ✓	No ✓
4×5	4	No	Yes

4) [10pts] Find the inverse matrix of the invertible matrix $A = \begin{bmatrix} 1 & -3 & -1 \\ 0 & 4 & 5 \\ 1 & -2 & 0 \end{bmatrix}$ and check your answer by performing matrix multiplication. [Hint: all of the entries of the inverse matrix are integers]

$$\begin{bmatrix} 1 & -3 & -1 & 1 & 0 & 0 \\ 0 & 4 & 5 & 0 & 1 & 0 \\ 1 & -2 & 0 & 0 & 0 & 1 \end{bmatrix} \quad (A)A^{-1} = \begin{bmatrix} 1 & -3 & -1 \\ 0 & 4 & 5 \\ 1 & -2 & 0 \end{bmatrix} \begin{bmatrix} -10 & -2 & 11 \\ -5 & -1 & 5 \\ 4 & 1 & -4 \end{bmatrix} \quad A^{-1} = \begin{bmatrix} -10 & -2 & 11 \\ -5 & -1 & 5 \\ 4 & 1 & -4 \end{bmatrix}$$

$$\underbrace{R_3 \rightarrow R_3 - R_1}_{\sim} \begin{bmatrix} 1 & -3 & -1 & 1 & 0 & 0 \\ 0 & 4 & 5 & 0 & 1 & 0 \\ 0 & 1 & 1 & -1 & 0 & 1 \end{bmatrix}$$

$$\underbrace{R_2 \rightarrow \frac{R_2}{4}}_{\sim} \begin{bmatrix} 1 & -3 & -1 & 1 & 0 & 0 \\ 0 & 1 & 5/4 & 0 & 1/4 & 0 \\ 0 & 1 & 1 & -1 & 0 & 1 \end{bmatrix}$$

$$\underbrace{R_3 \rightarrow R_3 - R_2}_{\sim} \begin{bmatrix} 1 & -3 & -1 & 1 & 0 & 0 \\ 0 & 1 & 5/4 & 0 & 1/4 & 0 \\ 0 & 0 & -1/4 & -1 & -1/4 & 1 \end{bmatrix}$$

$$\underbrace{R_3 \rightarrow -4R_3}_{\sim} \begin{bmatrix} 1 & -3 & -1 & 1 & 0 & 0 \\ 0 & 1 & 5/4 & 0 & 1/4 & 0 \\ 0 & 0 & 1 & 4 & 1 & -4 \end{bmatrix}$$

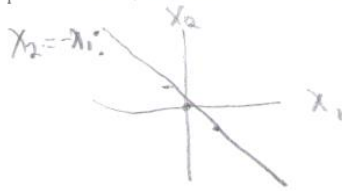
$$\underbrace{R_2 \rightarrow R_2 - \frac{5}{4}R_3}_{\sim} \begin{bmatrix} 1 & -3 & -1 & 1 & 0 & 0 \\ 0 & 1 & 0 & -5 & -1 & 5 \\ 0 & 0 & 1 & 4 & 1 & -4 \end{bmatrix}$$

$$\underbrace{R_1 \rightarrow R_1 + R_3}_{\sim} \begin{bmatrix} 1 & -3 & 0 & 5 & 1 & -4 \\ 0 & 1 & 0 & -5 & -1 & 5 \\ 0 & 0 & 1 & 4 & 1 & -4 \end{bmatrix}$$

$$\underbrace{R_1 \rightarrow R_1 + 3R_2}_{\sim} \begin{bmatrix} 1 & 0 & 0 & -10 & -2 & 11 \\ 0 & 1 & 0 & -5 & -1 & 5 \\ 0 & 0 & 1 & 4 & 1 & -4 \end{bmatrix}$$

5) [14pts] Let T and S be linear transformations from $\mathbb{R}^2$ to $\mathbb{R}^2$. T maps $\mathbf{e}_1$ to $3\mathbf{e}_1$ and $\mathbf{e}_2$ to $\mathbf{e}_1 - 5\mathbf{e}_2$ and S reflects points through the line $x_2 = -x_1$. Let A , B , and C denote the standard matrices of T , S , and $S \circ T$, respectively. Find A , B , and C .

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$



$$A = \begin{bmatrix} 3 & 1 \\ 0 & -5 \end{bmatrix}$$

$$T(3\mathbf{e}_1) = 3 \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$$

$$T(\mathbf{e}_1 - 5\mathbf{e}_2) = \begin{bmatrix} 1 \\ 0 \end{bmatrix} - 5 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 5 \end{bmatrix} = \begin{bmatrix} 1 \\ -5 \end{bmatrix}$$

$$B = \begin{bmatrix} -3 & 1 \\ 0 & -5 \end{bmatrix}$$

$$C = \begin{bmatrix} -9 & -2 \\ 0 & 25 \end{bmatrix}$$

$$S \circ T = ? \rightarrow \begin{bmatrix} 3 & 1 \\ 0 & -5 \end{bmatrix} \begin{bmatrix} -3 & 1 \\ 0 & -5 \end{bmatrix} = \begin{bmatrix} -9 & -2 \\ 0 & 25 \end{bmatrix}$$

6) [8pts] Let $A = \begin{bmatrix} 1 & 5 & 3 \\ 2 & 7 & 8 \\ 0 & -3 & 2 \\ 1 & 2 & 5 \end{bmatrix}$ and $T(\mathbf{x}) = A\mathbf{x}$. Is T a one-to-one transformation? Why or why not?

Show your work and circle one of the following statements.

- a) Yes, T is one-to-one because the columns of A span $\mathbb{R}^3$.
- b) Yes, T is one-to-one because the columns of A span $\mathbb{R}^4$.
- c) Yes, T is one-to-one because none of the columns of A is a constant multiple of another.
- d) Yes, T is one-to-one because the equation $A\mathbf{x} = \mathbf{b}$ has a solution for every $\mathbf{b}$ in $\mathbb{R}^4$.
- e) Yes, T is one-to-one because the equation $A\mathbf{x} = \mathbf{0}$ has a free variable.
- f) No, T is not one-to-one because the columns of A are linearly independent.
- g) No, T is not one-to-one because the columns of A are linearly dependent.
- h) No, T is not one-to-one because the equation $A\mathbf{x} = \mathbf{0}$ has no free variable.
- i) No, T is not one-to-one because the equation $A\mathbf{x} = \mathbf{b}$ is inconsistent for every $\mathbf{b}$ in $\mathbb{R}^4$.
- j) No, T is not one-to-one because the equation $A\mathbf{x} = \mathbf{0}$ does not have the trivial solution.

$$A \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \begin{bmatrix} 1 & 5 & 3 \\ 0 & -3 & 2 \\ 0 & -3 & 2 \\ 1 & 2 & 5 \end{bmatrix} \quad \text{FF?} \quad (-3)$$

Since R_2 & R_3 are multiples of each other $\rightarrow$ columns of A are linearly dependent.

7) Consider the matrix $A = \begin{bmatrix} 1 & -4 \\ -2m & 3m^2 \end{bmatrix}$.

a) [6pts] For all values of m for which A is invertible, find A^{-1} .

$$\det A = (1)(3m^2) - (-4)(-2m) \\ = 3m^2 - 8m$$

$$A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$A^{-1} = \frac{1}{3m^2 - 8m} \begin{bmatrix} 3m^2 & 4 \\ 2m & 1 \end{bmatrix}$$

b) [4pts] What are the values of m mentioned in part (a)? That is, for which values of m does the inverse of A exist?

$$3m^2 - 8m = 0 \Rightarrow m(3m - 8) = 0$$

$$3m - 8 = 0 \Rightarrow m = 8/3$$

$$A^{-1} \text{ exists for all values except } m=0 \text{ \& } m = \frac{8}{3}$$

8) [8pts] Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a linear transformation given by

$$T \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \right) = \begin{bmatrix} 3x_1 \\ -9x_1 + 4x_2 \\ 10x_1 + 17x_2 - 5x_3 \end{bmatrix}, \text{ for all } \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \text{ in } \mathbb{R}^3.$$

Let P be a parallelepiped in $\mathbb{R}^3$. If the volume of P is 15 (cubic units), find the volume of $T(P)$, where $T(P)$ is the image of P under T .

$$A = \begin{bmatrix} 3 & 0 & 0 \\ -9 & 4 & 0 \\ 10 & 17 & -5 \end{bmatrix} \Rightarrow \det(A) = \overbrace{(3)(4)(-5)}^{\text{By Triangle Matrix}} = -60$$

$$\begin{array}{r} 60 \\ 15 \\ \hline 300 \\ 600 \\ \hline 900 \end{array}$$

$$\text{Volume of } T(P) = |\det(A)| (\text{Volume of } P) = (-60)(15) = \boxed{-900}$$

(-1)

9) For all parts of this problem, A , B , and C denote matrices.

[2pts each] Determine whether each statement is **True** or **False**. Circle or box your answer. No explanation is necessary.

(-4)

a) Suppose AB and AC are defined and $A \neq O$. If $AB = AC$, then $B = C$.

True **False**

b) $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

True **False**

c) If A and B are 3×3 matrices and $B = [\mathbf{b}_1 \ \mathbf{b}_2 \ \mathbf{b}_3]$, then $AB = [A\mathbf{b}_1 + A\mathbf{b}_2 + A\mathbf{b}_3]$.

True False

d) Suppose A has size $n \times n$. If the transformation $\mathbf{x} \mapsto A\mathbf{x}$ is onto, then the transformation $\mathbf{x} \mapsto A^T \mathbf{x}$ is invertible.

True False

e) If the columns of an $n \times n$ matrix are linearly dependent, then its rows are also linearly dependent.

True **False**

10) [10pts] Consider the linear transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ given by:

$$T \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \right) = \begin{bmatrix} x_1 - x_3 \\ -2x_1 + 3x_2 + x_3 \\ 3x_2 - x_3 \end{bmatrix}, \text{ for all } \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \text{ in } \mathbb{R}^3.$$

a) Find the standard matrix of the linear transformation T and reduce it to an echelon form.

Standard Matrix $A = \begin{bmatrix} 1 & 0 & -1 \\ -2 & 3 & 1 \\ 0 & 3 & -1 \end{bmatrix} \xrightarrow{R_2 \rightarrow R_2 + 2R_1} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 3 & -1 \\ 0 & 3 & -1 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_2} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 3 & -1 \\ 0 & 0 & 0 \end{bmatrix}$ E.F.

b) Is T an invertible transformation? Why or why not?

Circle one: Yes **No**

Explain.

$\det A = 0 \rightarrow$ Since $\det A = 0$, A is not invertible.

Since A is not invertible, T is not an invertible transformation.